

Power Analysis and Experiment Design

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1 Introduction

This document details the design choices for the comparative experiment between a synthetic Fitts' law experiment with distractors, and a Fitts' law experiment using realistic targets and scenes. The synthetic experiment is essentially a replication of the experiment by [1], although it is not a replication in the strict sense, since we don't try to match conditions perfectly.

2 Choice of conditions

The main factors are:

- **TECHNIQUE**: Discrete factor with 5 values *Regular Pointing*, *Semantic Pointing*, *Bubble-Cursor*, *DynaSpot*, *RakeCursor*
- **ID**: Continuous factor, with 4 levels manipulated in the experiment: $ID \in [2, 4, 6, 8]$.
- **DENSITY**: Continuous factor, with 3 levels manipulated in the experiment: $\rho \in [0.1, 0.3, 0.6]$

IDs were chosen to span the common ID range encountered in interfaces. The values for ρ represent very dense to very sparse environments. A full factorial design the aforementioned factors yields 60 conditions. Considering about 7 seconds per average for each movement (estimated from personal tests, this includes breaks and time between conditions), one trial per condition should result in an experiment that lasts about 7 minutes, which, if we consider a total budget of 45-60 minutes, makes it possible to have between 7 and 9 trials at maximum per condition and per participant.

3 Data Generating Process

To determine the number of participants required for this experiment, we first produced a complete data generating process (DGP) that was fit to the data reported in [1]:

- We used the available data summary presented in [1] as reference, from now on referred to as BO data
- We assumed a linear mixed effect model as data generating process for the BO data:

$$MT = \alpha + \alpha_i + \beta_j + (\gamma + \gamma_i + \gamma_j)ID + (\omega + \omega_i + \omega_j)\rho + \varepsilon_{ijk}, \quad (1)$$

This corresponds to the following statistical formula, expressed in R-syntax:

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MT ~ 1 + ID * Technique + Technique * Density + (1+ ID + Density||Participants)
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where i indexes participants, j indexes techniques, and k indexes trials. This model captures the following effects:

- A global average movement time (α)
 - Difference between participants in average movement time (α_i)
 - Difference between techniques in average movement time (α_j)
 - An average effect of ID (γ , slope) and different slopes per participant (γ_i) and per technique (γ_j).
 - An average effect of density ρ (ω), and different effects of density per participant (ω_i) and per technique (ω_j)
 - Residual variability (ε_{ijk}).
- All terms indexed by i are realizations of zero-mean Gaussian random variables that represent inter-participant variability, for which we have to specify the variances. To do so, we assumed that the participant variability was such that 96% (*i.e.*, $\sim 4\sigma$) of the realized data would be contained within 20 percent of the associated main effect. For example, for a slope computed to be $b = 1\text{bit/s}$, we chose the standard deviation for γ_i as $\sigma = \frac{1 \times 0.2}{4}$.
 - The terms indexed by j are fixed effects, and refer to the specific slopes and density effect for each technique. These were estimated from the individual fits reported in [1].
 - To estimate the variance of the only remaining term (the residuals, ε_{ijk}), we used Equation 1 to generate data for the same number of participants and trials as in the BO data. The standard deviation of the residuals was then fit so that the summary statistics of the dataset (means and standard deviations of MT for each technique) matched as best as possible those reported in [1]. This resulted in a value of $\sigma \sim 10^{-0.8}$. Fitting on reduced statistics is similar in spirit to approximate Bayesian computation as presented in [2], although we don't use a Bayesian approach here for simplicity.

4 Sample size determination

Equipped with a fully specified DGP, we ran a Monte Carlo power analysis [3], which answers the following questions: *For this design, how many of my fixed effects are well-powered?* We thus computed the number of fixed effects for the model in Equation 1 that attain a power of at least 80% for a significance of 5%, for varying number of participants (N) and trials (n).

The result is given Figure 1.

5 Counterbalancing

Proper counterbalancing to remove 1st and 2nd order effects would require multiples of 60 participants using a balanced latin square (BLS), but the previous analysis shows we don't require as many. Thus, we need to find a counterbalancing strategy that is adequate for about 20 participants. The most important condition to counterbalance is TECHNIQUE, because it is likely to have strong learning or transfer effects, compared to ID and DENSITY, which are only "stimulus" properties, and which have likely limited order effects. Thus, we only counterbalance TECHNIQUE using a BLS, which requires multiple of 10 participants. This defines blocks at the technique level. Within each of these blocks, all 12 conditions will be performed by all participants, but the order of these

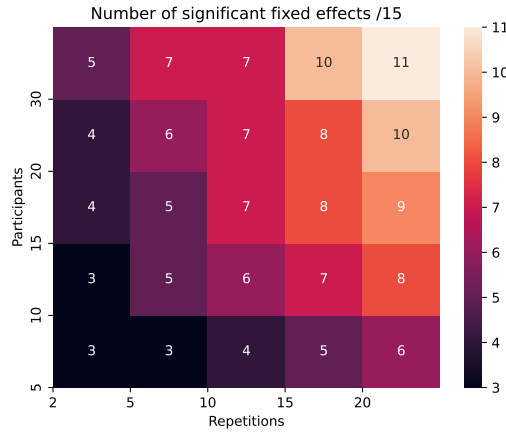


Figure 1: Tradeoff between number of repetitions per condition and number of participants in terms of statistical power.

conditions will be randomized (concretely, we repeatedly sample without replacement conditions from the initial pool of 12 conditions).

6 Limitations

The suitability of the Monte Carlo power analysis is primarily dependent on the soundness of the DGP. Several choices could have been made differently. For example: The DGP treats the random slopes as uncorrelated: $(1 + \text{ID} + \text{Density} | \text{Participants})$, but they are likely correlated: $(1 + \text{ID} + \text{Density} | \text{Participants})$. This choice was made for simplicity, as this avoids specifying covariance parameters which would be impossible to determine without the full dataset. This may overestimate power a bit.

The major modeling choice is that fixed interaction effects were not included, such as $\text{ID} \times \text{DENSITY}$. This leads to a simpler model with fewer parameters, and thus easier to identify, but we lose the ability to discuss possible interactions between density and ID (which assumes that motor difficulty and the effect of distractors are additive), or between ID and technique (which assumes that techniques work equally well for small and large targets). Another reason for not considering these is that these interactions terms can not be estimated from the data reported in [1].

References

- [1] Renaud Blanch and Michael Ortega. Benchmarking pointing techniques with distractors: adding a density factor to fitts’ pointing paradigm. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 1629–1638, 2011.
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- [3] Linda K Muthén and Bengt O Muthén. How to use a monte carlo study to decide on sample size and determine power. *Structural equation modeling*, 9(4):599–620, 2002.